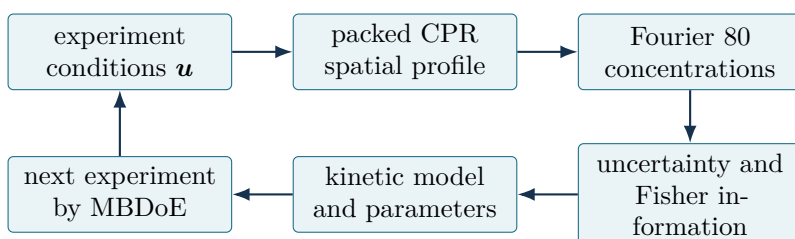

From Spatial Profiles to Autonomous Experiment Selection

A Comprehensive Theory and Layer-2 Blueprint
for the CPR–Flow-NMR Self-Driving Laboratory



Prepared as a unified teaching document from the complete project discussion

July 2026

This document combines the physical interpretation, worked examples, matrix theory, autonomous design logic, validation strategy, and implementation plan without repeating the same ideas in separate forms.

Purpose and reading guide

This document answers one central question:

How can a spatially sampled plug-flow reactor, coupled to a benchtop NMR, learn a kinetic model autonomously and decide which operating condition should be tested next?

The answer is developed in the order in which the ideas logically depend on each other. The physical reactor and measurement system come first. The analytical $A \rightarrow B \rightarrow C$ example then makes spatial information tangible. Only after that are the measurement vector, covariance, sensitivity matrix, Fisher Information Matrix, parameter covariance, and model-based design of experiments constructed. The final chapters translate the mathematics into a closed-loop Layer-2 software plan, validation benchmark, test strategy, and definition of done.

Two uses of the word *layer* appeared during the project discussion. At the complete platform level it is useful to distinguish (i) reactor physics, (ii) the measurement chain, and (iii) inference and decision making. In the software implementation plan, however, the existing PFR simulator is called *Layer 1*, and the new observation, estimation, information, and autonomous-design framework is called *Layer 2*. This document uses the software convention while explicitly separating the measurement and decision functions inside Layer 2.

The mathematical backbone

The primary deliverable is not merely a parameter-fitting loop. It is a transparent framework that explicitly constructs

$$\mathbf{Y} \longrightarrow \mathbf{y} \longrightarrow \hat{\mathbf{y}} \longrightarrow \mathbf{r} \longrightarrow \mathbf{S} \longrightarrow \mathbf{F} \longrightarrow \mathbf{V}_\phi \longrightarrow \mathbf{u}_{\text{next}},$$

retains labels and units for every row and column, and uses these objects to determine the next experiment.

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What “autonomous kinetic modelling” actually means

1.1 The apparent paradox

For a simple reaction, it is tempting to think that kinetic modelling means choosing a rate law, fitting one rate constant k , and repeating the fit at several temperatures to obtain an Arrhenius relation. If that were the complete problem, there would be little need for a self-driving laboratory. One could simply choose a few temperatures, measure conversion, and regress $k(T)$.

Real systems are harder for three reasons. First, several parameter combinations can produce similar outlet conversions. Second, a reversible or consecutive network contains multiple forward and reverse processes whose influence changes with temperature and composition. Third, the measured signal is not the reactor state itself: hydrodynamics, sampling delay, line dispersion, ongoing reaction, spectral overlap, and calibration uncertainty lie between the chemical state and the concentration reported by the NMR.

The autonomous problem is therefore not:

“Which experiment gives the highest yield?”

It is initially:

“Given what has already been measured, which safe experiment will most reduce the remaining uncertainty in the kinetic model?”

Optimization of yield, selectivity, energy use, or economics can follow after a sufficiently predictive model has been learned. Mixing model identification and process optimization too early can produce a condition that looks optimal only because the model is wrong or poorly identified.

1.2 What counts as the next experiment?

An experiment is one complete steady operating condition,

$$\mathbf{u}_e = \begin{bmatrix} T_e & Q_e & C_{\text{EGDA},0,e} & C_{\text{H}_2\text{O},0,e} & C_{\text{AcOH},0,e} & C_{\text{cat},e} & \cdots \end{bmatrix}^T.$$

The exact active entries depend on the pumps, feeds, chemistry, and Layer-1 simulator. The next experiment might change:

- reactor temperature;
- total volumetric flow rate and therefore the spatial reaction-age range;
- inlet EGDA concentration;
- water-to-ester ratio;
- acid catalyst concentration;
- initial AcOH concentration, which can excite reverse-reaction effects;
- a supported solvent or composition variable;
- in a later implementation, the commanded axial sampling positions or measurement allocation.

Moving the capillary orifice to a new axial position is not a new reactor experiment. The positions are multiple observations within the same operating condition. This distinction is important when counting experimental effort.

1.3 The closed learning loop

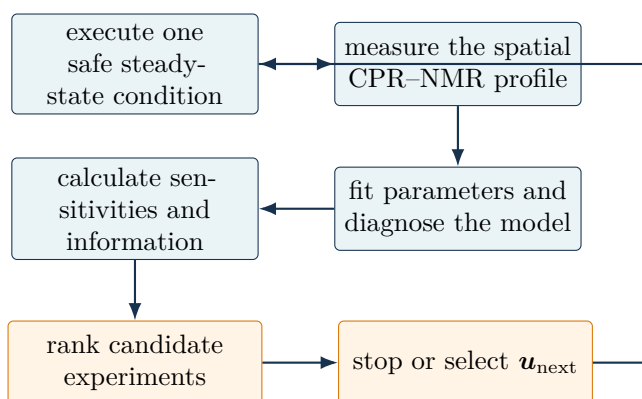


Figure 1.1: The autonomous campaign is a sequential learning loop, not a one-shot optimizer.

After every experiment the system updates both the parameter estimate and the uncertainty. Candidate conditions are simulated using only the current estimate or its uncertainty distribution. The hidden truth used in a virtual campaign must never be visible to the estimator or experiment selector.

The physical platform and the meaning of a spatial profile

2.1 Hardware concept

The proposed proof-of-concept combines:

1. a Reacnostics Catalytic Profiling Reactor (CPR);
2. an inert bed of glass, silica, or quartz particles;
3. a homogeneous liquid-phase reaction and liquid acid catalyst;
4. one axially movable sampling capillary with a sampling orifice;
5. one direct transfer tube from that capillary to a Bruker Fourier 80 with an InsightMR flow cell;
6. continuous-flow NMR acquisition and quantitative concentration inference;
7. a Layer-1 reactor model and a Layer-2 estimation/MBDoE controller.

The beads do not catalyse the reaction. Their purposes are hydrodynamic: disrupt the empty-tube laminar velocity profile, create a reproducible interstitial flow field, and mechanically support the movable sampling capillary. The chemistry occurs in the liquid void space.

2.2 Reaction initiation and the hydraulic origin

The reactants and catalyst should remain separated until a low-holdup micromixer immediately upstream of the CPR. This defines a meaningful origin $z = 0$ and reaction age $t = 0$. Reaction in long upstream feed lines would otherwise add an uncertain, condition-dependent prehistory.

The preferred downstream sequence is:

movable capillary orifice at $z \rightarrow$ sampling capillary $\rightarrow$
direct transfer tube $\rightarrow$ InsightMR cell $\rightarrow$ continuous-flow spectrum.

There is no selector, valve manifold, or set of separate branches. The capillary itself selects the reactor coordinate by moving its orifice along the axis.

2.3 Position is primary; residence time is derived

The Layer-1 model is a steady spatial model. Its independent coordinate is the axial position z , not an independently imposed transient time. For an ideal incompressible packed PFR with constant void fraction ε_b , cross-sectional area A_r , and volumetric flow Q , the nominal reaction age at position z is

$$\tau(z) = \frac{\varepsilon_b A_r z}{Q}.$$

This mapping is useful for interpretation, but it is derived from the hydrodynamics. It is not a second experimental coordinate. The capillary may be positioned continuously over its accessible axial range, but a measured profile is normally represented by a finite set of commanded positions $z_1, \dots, z_{N_z}$. These are repeated observations from one steady spatial field, not separate transient experiments and not separate physical ports.

Do not relabel the model

The spatial PFR should be solved in z . A residence-time axis may be plotted afterward when the mapping is justified. Treating τ as an independent control variable can silently duplicate information or hide hydrodynamic assumptions.

2.4 Porosity and the packed homogeneous PFR balance

Define the superficial and interstitial velocities as

$$u_s = \frac{Q}{A_r}, \quad u_{\text{int}} = \frac{u_s}{\varepsilon_b}.$$

If the reaction-rate vector $\mathbf{R}$ is expressed per interstitial liquid volume, the steady ideal PFR balance is

$$u_{\text{int}} \frac{d\mathbf{C}}{dz} = \mathbf{R}(\mathbf{C}, T, \boldsymbol{\theta}),$$

or equivalently,

$$u_s \frac{d\mathbf{C}}{dz} = \varepsilon_b \mathbf{R}(\mathbf{C}, T, \boldsymbol{\theta}).$$

Omitting ε_b while using superficial velocity overestimates the liquid residence time and biases fitted kinetic constants.

The sampling flow q_s is a sidestream withdrawn only at the current orifice position z_j . For an ideal PFR, a composition-proportional withdrawal at z_j does not change the concentrations immediately upstream of the orifice, so the prediction assigned to that position is still obtained by integrating from 0 to z_j using the inlet flow Q_{in} . If the downstream reactor state during sampling must also be represented, the flow is piecewise:

$$Q(z \mid z_j) = Q_{\text{in}} - q_s H(z - z_j),$$

where H is the Heaviside step function. The corresponding nominal reaction age is then

$$\tau(z \mid z_j) = \int_0^z \frac{\varepsilon_b A_r}{Q(\zeta \mid z_j)} d\zeta.$$

When q_s/Q_{in} is negligible, the original constant- Q PFR is the appropriate non-invasive approximation. If the withdrawal is not negligible, or axial dispersion transmits the disturbance around the orifice, the position-dependent sidestream must be included and the profile must be allowed to re-establish after capillary motion.

For the nominal example discussed previously, a 4 mm internal diameter and 60 mm length give a geometric bed volume of approximately 0.754 mL. With $\varepsilon_b = 0.40$, the reacting liquid holdup is approximately 0.30 mL. Nominal whole-bed reaction ages are then:

Flow rate	Liquid reaction age
1.0 mL.min ⁻¹	18 s
0.5 mL.min ⁻¹	36 s
0.1 mL.min ⁻¹	180 s

2.5 Plug flow is a qualification result, not an assumption

Inert packing usually improves radial mixing relative to an empty laminar tube, but it does not guarantee ideal plug flow. Channeling, wall-induced voidage gradients, stagnant zones, and axial dispersion can still occur. A first extension is the axial-dispersion model

$$\mathbf{0} = D_{\text{ax}} \frac{d^2 \mathbf{C}}{dz^2} - u_{\text{int}} \frac{d\mathbf{C}}{dz} + \mathbf{R}(\mathbf{C}, T, \boldsymbol{\theta}),$$

with appropriate Danckwerts boundary conditions. The Bodenstein number

$$\text{Bo} = \frac{u_{\text{int}} L}{D_{\text{ax}}}$$

quantifies the relative importance of convection and dispersion. The ideal PFR should be retained only if tracer RTD data and model comparison support it over the intended operating range.

A practical starting particle-size range for a 4 mm tube is roughly 0.4–0.6 mm, corresponding to a tube-to-particle ratio near 7–10. This is an engineering starting point, not a universal optimum. Pressure drop, RTD, wall effects, and repacking reproducibility must be measured.

2.6 The sampling path is part of the observation model

For cylindrical tubing of length L_t and internal diameter d ,

$$V_t = \frac{\pi d^2}{4} L_t, \quad \tau_t = \frac{V_t}{q}.$$

For 10 cm of tubing, the approximate internal volumes and delays at 0.5 mL.min⁻¹ are:

Internal diameter	Internal volume	Nominal delay
100 μm	0.79 μL	0.09 s
250 μm	4.9 μL	0.59 s
1 mm	78.5 μL	9.4 s

The quadratic dependence on diameter makes both the sampling capillary and the direct transfer tube important. Their internal diameters and lengths should be kept as small as is compatible with reliable withdrawal, pressure drop, and continuous NMR flow.

For a commanded position z_j , the nominal mean transport delay to the centre of the NMR sensitive region may be approximated as

$$\tau_d(z_j) = \frac{V_{\text{cap}}(z_j) + V_{\text{tube}} + V_{\text{cell,pre}} + \frac{1}{2} V_{\text{sens}}}{q_s}.$$

Here V_{cap} is the liquid volume from the sampling orifice to the capillary outlet, V_{tube} is the single direct tube volume, $V_{\text{cell,pre}}$ is the cell volume before the sensitive region, V_{sens} is the NMR-sensitive liquid volume, and q_s is the continuous sampling flow. The factor $\frac{1}{2}V_{\text{sens}}$ locates the nominal signal at the centre of the sensitive volume; the real weighting should be measured because velocity profiles and dispersion broaden it. If a spectrum is timestamped at the start rather than at the midpoint of its acquisition window, approximately half the acquisition time must also be included when aligning it with the capillary position.

For a rigid capillary whose orifice-to-outlet length is fixed, V_{cap} is independent of z_j , so the nominal hydraulic delay is the same at every axial position. Position-dependent delays should be introduced only if motion actually changes the wetted path volume or flow.

Because the acid-catalysed reaction can continue after withdrawal, the transfer path may need a short reactive-PFR or dispersed-delay submodel. Physical mitigations include small capillary and transfer-tube volume, sufficient sampling flow, cooling where compatible with quantitation, and measured line RTDs. After moving from z_{j-1} to z_j , the single path must be flushed or allowed to wash out before the spectrum is assigned to z_j ; there are no unsampled branches containing stale liquid.

The sampling withdrawal should be a small fraction of the main flow. A nominal starting target below roughly 1–2% is sensible, but the decisive test is profile invariance as the withdrawal rate is changed.

Continuous positioning versus discrete observations

The capillary is continuously positionable in z , but the matrix model below assumes step-and-hold positioning with continuous sample flow: move to z_j , flush and stabilize the one path, then acquire the flowing NMR signal. If spectra are acquired while the capillary is moving, each datum is a convolution over the trajectory $z(t)$, the acquisition window, the NMR-sensitive volume, and the transfer-path residence-time distribution; it can no longer be identified with a single $C_i(z_j)$ without an additional scanning observation model.

2.7 The actual measurement operator

The NMR output is not automatically the instantaneous reactor concentration at the capillary orifice. A more faithful observation model is

$$y_{j,i} = g_{\text{NMR},i} [\mathcal{M}_{\text{path}}(\mathbf{C}(z_j), T_{\text{path}}, \tau_d(z_j), \boldsymbol{\theta})] + \varepsilon_{j,i},$$

where the same operator $\mathcal{M}_{\text{path}}$ represents delay, dispersion, washout, and continued reaction in the one capillary-tube-cell path, while $g_{\text{NMR},i}$ maps the resulting mixture and spectrum to a concentration estimate.

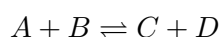
At 80 MHz, peak overlap, baseline drift, water suppression, temperature-dependent shifts, and common calibration errors can create correlated concentration errors. Constrained spectral fitting or indirect hard modelling is therefore preferable to blind integration of every peak. The concentration estimator should return both concentrations and uncertainty information.

Reaction models: from a starter reaction to EGDA

3.1 Why begin with a simple ester hydrolysis?

Ethyl acetate hydrolysis is a useful commissioning reaction because its chemistry, NMR signals, and expected trends are comparatively easy to understand. It can be used to test pumps, mixing, capillary positioning, path washout, transfer delay, NMR quantitation, and a one-step kinetic fit before the full consecutive reversible model is activated.

A generic reversible hydrolysis is



with a concentration-based rate

$$r = k_f C_A C_B - k_r C_C C_D.$$

An activity-based form is

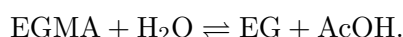
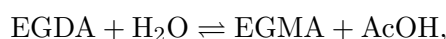
$$r = k_f a_A a_B - k_r a_C a_D = k_f a_A a_B \left(1 - \frac{Q_a}{K_a}\right),$$

where $Q_a = a_C a_D / (a_A a_B)$ and $K_a = k_f / k_r$ under the chosen consistent standard-state convention.

If water is in sufficiently large excess, its activity changes little and the reaction can appear pseudo-first-order. That simplification should be tested, not assumed, especially when the project goal is to identify reversible, composition-dependent kinetics.

3.2 The EGDA network

The main use case is sequential reversible hydrolysis:



Using $A = \text{EGDA}$, $B = \text{water}$, $C = \text{EGMA}$, $D = \text{AcOH}$, and $E = \text{EG}$, one general activity-based model is

$$r_1 = k_{1f} a_A a_B a_{\text{H}^+}^{m_1} - k_{1r} a_C a_D a_{\text{H}^+}^{n_1}, \quad (3.1)$$

$$r_2 = k_{2f} a_C a_B a_{\text{H}^+}^{m_2} - k_{2r} a_E a_D a_{\text{H}^+}^{n_2}. \quad (3.2)$$

The stoichiometric source terms are

$$\mathbf{R} = \begin{bmatrix} -r_1 \\ -r_1 - r_2 \\ r_1 - r_2 \\ r_1 + r_2 \\ r_2 \end{bmatrix}.$$

The project should begin with the simplest scientifically defensible model: concentration-based rates and a fixed catalyst order if prior evidence supports it. Activity corrections, flexible orders, or additional mechanisms should be introduced only when validation residuals demonstrate that the simpler model is inadequate. Adding every plausible parameter at the beginning usually destroys identifiability.

3.3 Temperature dependence and parameterization

Raw Arrhenius parameters A_j and E_j are often strongly correlated. A more stable parameterization uses a reference temperature:

$$\ln k_j(T) = \alpha_j - \frac{E_j}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right), \quad \alpha_j = \ln k_j(T_{\text{ref}}).$$

Positive parameters should normally be optimized in transformed coordinates,

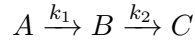
$$\boldsymbol{\phi} = \mathcal{T}(\boldsymbol{\theta}), \quad \phi_q = \ln \theta_q$$

for a strictly positive θ_q . All sensitivities and Fisher calculations must state whether they are with respect to physical parameters $\boldsymbol{\theta}$ or transformed parameters $\boldsymbol{\phi}$. This document uses $\boldsymbol{\phi}$ as the internal estimation coordinate unless stated otherwise.

Worked example: what spatial measurements reveal

4.1 Analytical consecutive reaction

Consider the irreversible network



with $C_A(0) = C_{A,0}$ and $C_B(0) = C_C(0) = 0$. Along reaction age τ ,

$$C_A(\tau) = C_{A,0}e^{-k_1\tau},$$

$$C_B(\tau) = C_{A,0} \frac{k_1}{k_1 - k_2} \left(e^{-k_2\tau} - e^{-k_1\tau} \right),$$

$$C_C(\tau) = C_{A,0} - C_A(\tau) - C_B(\tau).$$

Species B first forms and then disappears. Its internal maximum is precisely the feature that an outlet-only experiment can miss.

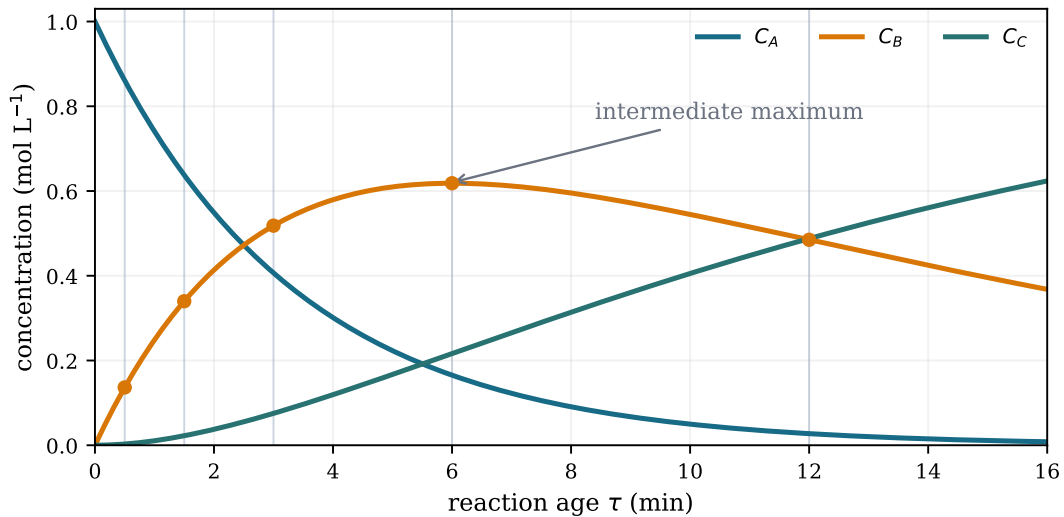


Figure 4.1: Analytical concentration profiles for $C_{A,0} = 1 \text{ mol.L}^{-1}$, $k_1 = 0.30 \text{ min}^{-1}$, and $k_2 = 0.08 \text{ min}^{-1}$. Vertical lines mark five illustrative CPR positions.

4.2 A concrete outlet ambiguity

Take two parameter sets:

$$\text{set 1: } (k_1, k_2) = (0.300, 0.080) \text{ min}^{-1},$$

$$\text{set 2: } (k_1, k_2) = (0.373, 0.077) \text{ min}^{-1}.$$

At $\tau = 12$ min their intermediate concentrations are almost identical:

$$C_B^{(1)} = 0.4849, \quad C_B^{(2)} = 0.4858 \text{ mol.L}^{-1}.$$

The reactant concentrations differ by only about 0.016 mol.L^{-1} , comparable to an illustrative measurement standard deviation of 0.020 mol.L^{-1} . Yet the internal profiles are visibly different.

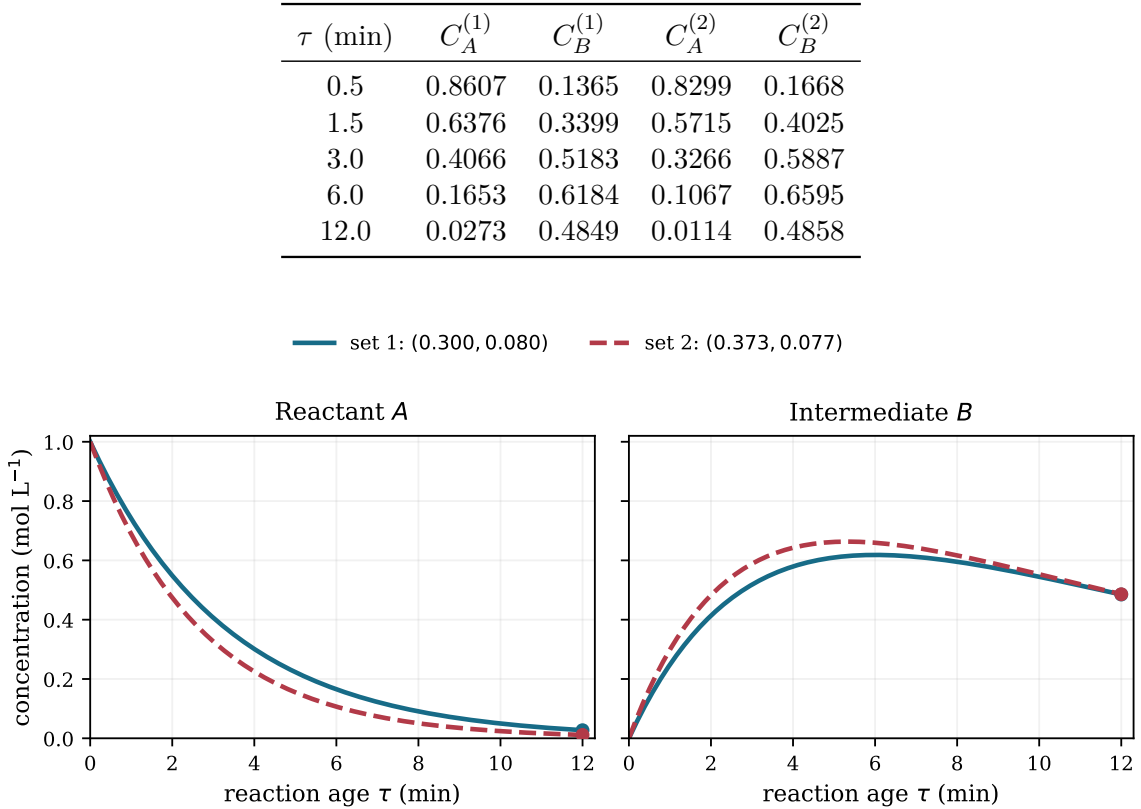


Figure 4.2: Two parameter sets can look almost identical at the outlet while producing different internal trajectories. Spatial observations break this ambiguity.

This example is deliberately simple. In the reversible EGDA network the ambiguity is stronger because forward rates, reverse rates, equilibrium effects, activities, acid dependence, and temperature dependence can compensate one another.

4.3 Analytical sensitivities

For measured species A and B ,

$$\frac{\partial C_A}{\partial k_1} = -\tau C_{A,0} e^{-k_1 \tau}, \quad \frac{\partial C_A}{\partial k_2} = 0.$$

Let $D = k_1 - k_2$ and $\Delta = e^{-k_2 \tau} - e^{-k_1 \tau}$. Then

$$\frac{\partial C_B}{\partial k_1} = C_{A,0} \left[-\frac{k_2}{D^2} \Delta + \frac{k_1}{D} \tau e^{-k_1 \tau} \right],$$

$$\frac{\partial C_B}{\partial k_2} = C_{A,0} \left[\frac{k_1}{D^2} \Delta - \frac{k_1}{D} \tau e^{-k_2 \tau} \right].$$

Early sampling positions contain strong information about disappearance of A and therefore k_1 . Later sampling positions accumulate information about consumption of B and therefore k_2 . The useful feature is not merely that there are more rows; their directions in parameter space differ.

The structured measurement matrix and observation vector

5.1 Indices and dimensions

Use:

$$e = 1, \dots, N_e, \quad j = 1, \dots, N_z^{(e)}, \quad i = 1, \dots, N_s^{(e)}, \quad q = 1, \dots, N_p.$$

Here e indexes reactor experiments, j sampling positions, i independently measured species, and q fitted parameters. Experiment e contains

$$m_e = N_z^{(e)} N_s^{(e)}$$

scalar observations. The framework must allow different experiments to use different sampling positions or measured species.

5.2 Structured matrix $\mathbf{Y}_e$

Arrange one observed spatial profile as

$$\mathbf{Y}_e = \begin{bmatrix} y_{e,1,1} & y_{e,1,2} & \cdots & y_{e,1,N_s} \\ y_{e,2,1} & y_{e,2,2} & \cdots & y_{e,2,N_s} \\ \vdots & \vdots & \ddots & \vdots \\ y_{e,N_z,1} & y_{e,N_z,2} & \cdots & y_{e,N_z,N_s} \end{bmatrix} \in \mathbb{R}^{N_z \times N_s},$$

where

$$y_{e,j,i} = C_i^{\text{obs}}(z_j; \mathbf{u}_e).$$

Rows are sampling positions and columns are independently quantified species. The prediction matrix is

$$\hat{\mathbf{Y}}_e(\boldsymbol{\phi}, \mathbf{u}_e) = \left[\hat{C}_i(z_j; \boldsymbol{\phi}, \mathbf{u}_e) \right]_{j,i}.$$

For a virtual experiment,

$$\mathbf{Y}_e = \hat{\mathbf{Y}}_e(\boldsymbol{\phi}_{\text{true}}, \mathbf{u}_e) + \mathbf{E}_e.$$

Only the virtual-reactor component may access $\boldsymbol{\phi}_{\text{true}}$.

5.3 One unambiguous position-first vector

At commanded axial position z_j define

$$\mathbf{y}_{e,j} = \begin{bmatrix} C_1^{\text{obs}}(z_j) \\ C_2^{\text{obs}}(z_j) \\ \vdots \\ C_{N_s}^{\text{obs}}(z_j) \end{bmatrix}.$$

The complete observation vector is

$$\mathbf{y}_e = \begin{bmatrix} \mathbf{y}_{e,1} \\ \mathbf{y}_{e,2} \\ \vdots \\ \mathbf{y}_{e,N_z} \end{bmatrix} \in \mathbb{R}^{m_e}.$$

The prediction vector must use the identical ordering:

$$\hat{\mathbf{y}}_e = \text{flatten}_{\text{position-first}} \left(\hat{\mathbf{Y}}_e \right).$$

Never rely on an undocumented array-flattening default. Every scalar row should carry

(experiment ID, sampling-position index, z , species, unit).

5.4 Five-position example

For two measured species and five sampling positions,

$$\mathbf{Y} = \begin{bmatrix} C_A(z_1) & C_B(z_1) \\ C_A(z_2) & C_B(z_2) \\ C_A(z_3) & C_B(z_3) \\ C_A(z_4) & C_B(z_4) \\ C_A(z_5) & C_B(z_5) \end{bmatrix} \in \mathbb{R}^{5 \times 2},$$

while

$$\mathbf{y} = \begin{bmatrix} C_A(z_1) \\ C_B(z_1) \\ C_A(z_2) \\ C_B(z_2) \\ C_A(z_3) \\ C_B(z_3) \\ C_A(z_4) \\ C_B(z_4) \\ C_A(z_5) \\ C_B(z_5) \end{bmatrix} \in \mathbb{R}^{10}.$$

This ordering must propagate through predictions, covariance matrices, residuals, sensitivity rows, plots, and saved files.

5.5 Independent measurements and mass balance

For $A \rightarrow B \rightarrow C$,

$$C_A + C_B + C_C = C_{A,0}.$$

If C_C is calculated as $C_{A,0} - C_A - C_B$, it is not an independent observation and must not be appended to $\mathbf{y}$ as if it created new experimental information. If all three species are independently quantified, all may be included, but their covariance must represent shared calibration and spectral effects. The same rule applies to EGDA: a mass-balance reconstruction is not a third or sixth NMR measurement.

Noise, covariance, residuals, and parameter estimation

6.1 Measurement covariance

Let

$$\boldsymbol{\varepsilon}_e = \mathbf{y}_e - \hat{\mathbf{y}}_e, \quad \boldsymbol{\varepsilon}_e \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{y,e}).$$

The covariance is

$$\boldsymbol{\Sigma}_{y,e} = \mathbb{E}[\boldsymbol{\varepsilon}_e \boldsymbol{\varepsilon}_e^T] \in \mathbb{R}^{m_e \times m_e}.$$

Its diagonal contains variances:

$$[\boldsymbol{\Sigma}_{y,e}]_{aa} = \sigma_a^2,$$

not standard deviations. Off-diagonal elements are error covariances.

Three useful levels of noise model are:

1. Equal independent errors:

$$\boldsymbol{\Sigma}_{y,e} = \sigma^2 \mathbf{I}.$$

For the five-position example, $\sigma = 0.020 \text{ mol.L}^{-1}$ gives $\boldsymbol{\Sigma}_y = (0.020)^2 \mathbf{I}_{10}$ in concentration-squared units.

2. Independent heteroscedastic errors:

$$\sigma_{e,j,i}^2 = \sigma_{i,\text{abs}}^2 + \left(\sigma_{i,\text{rel}} \hat{C}_i(z_j) \right)^2.$$

3. Correlated errors caused by overlapping peaks, common calibration, baseline drift, sequential position sampling, single-path memory, shared flow uncertainty, or spectral-model coupling.

Assuming independent equal noise when it is not justified makes the information matrix look artificially strong.

6.2 Raw and whitened residuals

Define

$$\mathbf{d}_e(\boldsymbol{\phi}) = \mathbf{y}_e - \hat{\mathbf{y}}_e(\boldsymbol{\phi}, \mathbf{u}_e).$$

If

$$\boldsymbol{\Sigma}_{y,e} = \mathbf{L}_e \mathbf{L}_e^T$$

is a Cholesky factorization, the whitened residual is

$$\mathbf{r}_e(\boldsymbol{\phi}) = \mathbf{L}_e^{-1} \mathbf{d}_e(\boldsymbol{\phi}).$$

The inverse is notation; the implementation should use a triangular solve.

Stacking all experiments gives

$$\mathbf{r}_{\text{all}} = \begin{bmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_{N_e} \end{bmatrix}.$$

The weighted nonlinear least-squares estimate is

$$\hat{\boldsymbol{\phi}} = \arg \min_{\boldsymbol{\phi} \in \Phi} \frac{1}{2} \mathbf{r}_{\text{all}}^{\text{T}} \mathbf{r}_{\text{all}}.$$

With a Gaussian prior $\mathcal{N}(\boldsymbol{\mu}_0, \mathbf{V}_0)$, the MAP objective is

$$\Phi_{\text{MAP}}(\boldsymbol{\phi}) = \frac{1}{2} \mathbf{r}_{\text{all}}^{\text{T}} \mathbf{r}_{\text{all}} + \frac{1}{2} (\boldsymbol{\phi} - \boldsymbol{\mu}_0)^{\text{T}} \mathbf{V}_0^{-1} (\boldsymbol{\phi} - \boldsymbol{\mu}_0).$$

6.3 Estimation is more than one optimizer call

A reliable estimator should include transformed parameters, physical bounds, multistart initialization, stable residual scaling, failed-simulation handling, and diagnostics. The required outputs are not merely best-fit values. They include residual plots, confidence intervals, correlations, numerical rank, profile likelihoods for suspect parameters, validation predictions, and flags for parameters that remain practically unidentifiable.

Sensitivity matrices: the bridge from physics to information

7.1 Definition

The sensitivity matrix is the Jacobian of model predictions, not a derivative of noisy measurements:

$$\mathbf{S}_e = \frac{\partial \hat{\mathbf{y}}_e}{\partial \boldsymbol{\phi}^\top} \in \mathbb{R}^{m_e \times N_p}.$$

Its element

$$[\mathbf{S}_e]_{a,q} = \frac{\partial \hat{y}_{e,a}}{\partial \phi_q}$$

states how one predicted scalar observation changes when one fitted parameter changes.

At one sampling position,

$$\mathbf{S}_{e,j} = \frac{\partial \hat{\mathbf{y}}_{e,j}}{\partial \boldsymbol{\phi}^\top} \in \mathbb{R}^{N_s \times N_p},$$

and the spatial sensitivity matrix is the vertical stack

$$\mathbf{S}_e = \begin{bmatrix} \mathbf{S}_{e,1} \\ \mathbf{S}_{e,2} \\ \vdots \\ \mathbf{S}_{e,N_z} \end{bmatrix}.$$

For the five-position, two-species, two-parameter example,

$$\mathbf{S}_{\text{spatial}} \in \mathbb{R}^{10 \times 2}, \quad \mathbf{S}_{\text{outlet}} = \mathbf{S}_5 \in \mathbb{R}^{2 \times 2}.$$

7.2 Numerical sensitivities

Central finite differences are the mandatory transparent baseline:

$$\frac{\partial \hat{\mathbf{y}}}{\partial \phi_q} \approx \frac{\hat{\mathbf{y}}(\boldsymbol{\phi} + h_q \mathbf{e}_q) - \hat{\mathbf{y}}(\boldsymbol{\phi} - h_q \mathbf{e}_q)}{2h_q}.$$

The implementation needs:

- scale-aware relative and absolute steps;

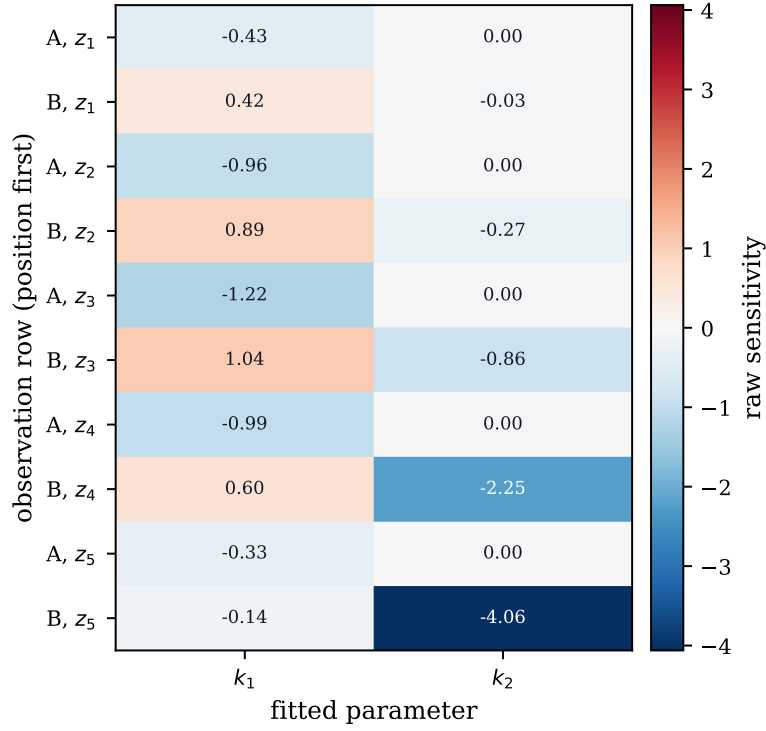


Figure 7.1: Raw position-first sensitivities for the five-position analytical example. Early rows strongly probe k_1 ; later B rows strongly probe k_2 .

- one-sided differences near bounds;
- tests across several step sizes;
- identical observation ordering in both perturbations;
- caching of repeated PFR simulations;
- graceful handling when one perturbation fails;
- comparison with analytical derivatives for $A \rightarrow B \rightarrow C$.

Automatic differentiation or integrated sensitivity equations can be added later, but they should be validated against the finite-difference baseline.

7.3 Whitened sensitivity

Define

$$\mathbf{J}_e = \mathbf{L}_e^{-1} \mathbf{S}_e.$$

$\mathbf{S}_e$ is the raw physical sensitivity; $\mathbf{J}_e$ is the sensitivity measured in units of observation noise. Both should be saved. Large raw sensitivity does not necessarily mean high information if the corresponding measurement is very noisy or strongly correlated.

Fisher information, identifiability, and parameter uncertainty

8.1 Information from one experiment

For experiment e ,

$$\mathbf{F}_e = \mathbf{S}_e^\top \Sigma_{y,e}^{-1} \mathbf{S}_e = \mathbf{J}_e^\top \mathbf{J}_e \in \mathbb{R}^{N_p \times N_p}.$$

The implementation should use linear solves or the whitened form, not an explicit covariance inverse.

With $\Sigma_y = \sigma^2 \mathbf{I}$,

$$\mathbf{F}_{\text{spatial}} = \frac{1}{\sigma^2} \mathbf{S}_{\text{spatial}}^\top \mathbf{S}_{\text{spatial}}, \quad \mathbf{F}_{\text{outlet}} = \frac{1}{\sigma^2} \mathbf{S}_5^\top \mathbf{S}_5.$$

8.2 Numerical result for the five-position example

For sampling positions at $\tau = (0.5, 1.5, 3, 6, 12)$ min and $\sigma = 0.020 \text{ mol.L}^{-1}$, the spatial and outlet-only information matrices are

$$\mathbf{F}_{\text{spatial}} \approx \begin{bmatrix} 15267.5 & -4834.2 \\ -4834.2 & 55961.6 \end{bmatrix},$$

$$\mathbf{F}_{\text{outlet}} \approx \begin{bmatrix} 318.2 & 1427.7 \\ 1427.7 & 41241.1 \end{bmatrix}.$$

Their approximate condition numbers are 3.85 and 154, respectively. The corresponding local parameter correlations are approximately +0.17 and -0.39. In this particular example the outlet still contains substantial information about k_2 , but it is weak in the k_1 direction; the internal sampling positions restore that missing direction.

8.3 When sampling-position information may be added

If errors at the commanded sampling positions are block independent,

$$\mathbf{F}_{\text{spatial}} = \sum_{j=1}^{N_z} \mathbf{S}_j^\top \Sigma_{y,j}^{-1} \mathbf{S}_j = \sum_{j=1}^{N_z} \mathbf{F}_j.$$

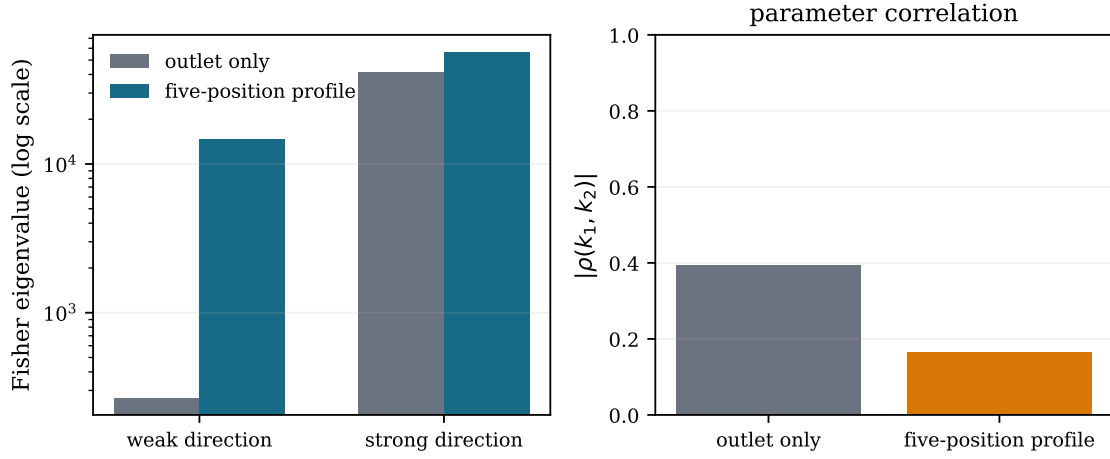


Figure 8.1: Spatial profiling raises the weak Fisher eigenvalue and reduces the magnitude of local parameter correlation in the worked example.

If errors across sampling positions are correlated, this additive calculation is generally invalid. One must construct the full covariance:

$$\mathbf{F}_{\text{spatial}} = \mathbf{S}_{\text{spatial}}^T \boldsymbol{\Sigma}_{\text{spatial}}^{-1} \mathbf{S}_{\text{spatial}}.$$

8.4 Stacking several reactor experiments

For N_e experiments,

$$\mathbf{y}_{\text{all}} = \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_{N_e} \end{bmatrix}, \quad \mathbf{S}_{\text{all}} = \begin{bmatrix} \mathbf{S}_1 \\ \vdots \\ \mathbf{S}_{N_e} \end{bmatrix},$$

with total length

$$M = \sum_{e=1}^{N_e} m_e.$$

The general accumulated data information is

$$\mathbf{F}_{\text{data}} = \mathbf{S}_{\text{all}}^T \boldsymbol{\Sigma}_{\text{all}}^{-1} \mathbf{S}_{\text{all}}.$$

If experiments are independent,

$$\boldsymbol{\Sigma}_{\text{all}} = \text{blockdiag}(\boldsymbol{\Sigma}_{y,1}, \dots, \boldsymbol{\Sigma}_{y,N_e})$$

and

$$\mathbf{F}_{\text{data}} = \sum_{e=1}^{N_e} \mathbf{F}_e.$$

Tests must demonstrate both this equality and its failure when cross-experiment correlation is introduced.

8.5 Prior information and total precision

If the prior covariance in transformed parameter space is $\mathbf{V}_0$, define

$$\mathbf{F}_{\text{prior}} = \mathbf{V}_0^{-1}, \quad \mathbf{P}_N = \mathbf{F}_{\text{prior}} + \mathbf{F}_{\text{data}}.$$

These are distinct objects:

- $\mathbf{F}_{\text{data}}$: information supplied by measurements;
- $\mathbf{F}_{\text{prior}}$: information supplied by prior knowledge;
- $\mathbf{P}_N$: approximate posterior precision.

Calling all three simply “the FIM” hides where the apparent certainty came from.

8.6 Covariance, standard deviations, and correlation

The local covariance approximation is

$$\mathbf{V}_\phi \approx \mathbf{P}_N^{-1}.$$

The inverse is again mathematical notation; stable factorization or solves should be used. For singular or nearly singular precision, report numerical rank, singular values, eigenvalues, condition number, and weak parameter combinations before considering a pseudoinverse.

If $\boldsymbol{\theta} = \mathcal{T}^{-1}(\phi)$ and

$$\mathbf{G} = \frac{\partial \boldsymbol{\theta}}{\partial \phi^\top},$$

then the delta method gives

$$\mathbf{V}_\theta \approx \mathbf{G} \mathbf{V}_\phi \mathbf{G}^\top.$$

Parameter standard deviations are

$$\sigma_{\theta_q} = \sqrt{[\mathbf{V}_\theta]_{qq}},$$

and with

$$\mathbf{D}_\theta = \text{diag}(\sigma_{\theta_1}, \dots, \sigma_{\theta_{N_p}}),$$

the correlation matrix is

$$\mathbf{R}_\theta = \mathbf{D}_\theta^{-1} \mathbf{V}_\theta \mathbf{D}_\theta^{-1}.$$

Large absolute off-diagonal values reveal parameter combinations that current experiments cannot separate.

8.7 Why adding internal sampling positions cannot reduce local information

Let $\mathbf{H}_{\text{out}}$ select the outlet rows:

$$\begin{aligned} \mathbf{y}_{\text{outlet}} &= \mathbf{H}_{\text{out}} \mathbf{y}_{\text{spatial}}, \\ \mathbf{S}_{\text{outlet}} &= \mathbf{H}_{\text{out}} \mathbf{S}_{\text{spatial}}, \quad \boldsymbol{\Sigma}_{\text{outlet}} = \mathbf{H}_{\text{out}} \boldsymbol{\Sigma}_{\text{spatial}} \mathbf{H}_{\text{out}}^\top. \end{aligned}$$

Under independent measurement errors,

$$\mathbf{F}_{\text{spatial}} = \mathbf{F}_{\text{outlet}} + \mathbf{F}_{\text{internal}}, \quad \mathbf{F}_{\text{internal}} \succeq \mathbf{0}.$$

Thus internal measurements cannot reduce local Fisher information under the assumed model. However, redundant rows may add almost nothing. The benefit comes from rows whose sensitivity directions are different from those already present.

What Fisher information does not prove

The FIM is local to a parameter point and a statistical/model assumption. A small local covariance does not prove global identifiability, model correctness, absence of bias, or predictive validity. It is a powerful design approximation, not a certificate of truth.

Choosing the next experiment

9.1 Candidate information

For every feasible candidate $\mathbf{u}_c$:

1. simulate the inference model at the current estimate;
2. construct $\hat{\mathbf{y}}_c$ in the defined position-first ordering;
3. calculate $\mathbf{S}_c$ with respect to the internal parameters ϕ ;
4. construct the predicted covariance $\Sigma_{y,c}$;
5. calculate

$$\mathbf{F}_c(\mathbf{u}_c) = \mathbf{S}_c^\top \Sigma_{y,c}^{-1} \mathbf{S}_c;$$

6. form the predicted updated precision

$$\mathbf{P}_{N+1}(\mathbf{u}_c) = \mathbf{P}_N + \mathbf{F}_c(\mathbf{u}_c).$$

The candidate calculation must use the current estimate or samples from the current approximate parameter distribution, never the hidden virtual truth.

9.2 D-, A-, and E-optimality

The default D-optimal information gain is

$$\Delta\Psi_D(\mathbf{u}_c) = \log \det[\mathbf{P}_N + \mathbf{F}_c(\mathbf{u}_c)] - \log \det(\mathbf{P}_N).$$

Select

$$\mathbf{u}_{\text{next}} = \arg \max_{\mathbf{u}_c \in \mathcal{U}_{\text{safe}}} \Delta\Psi_D(\mathbf{u}_c).$$

This approximately maximizes the contraction of the joint parameter-confidence ellipsoid.

Two alternatives are

$$\Psi_A(\mathbf{u}_c) = -\text{tr} [\mathbf{P}_{N+1}^{-1}(\mathbf{u}_c)],$$

which targets average marginal variance, and

$$\Psi_E(\mathbf{u}_c) = \lambda_{\min} [\mathbf{P}_{N+1}(\mathbf{u}_c)],$$

which strengthens the weakest identifiable direction.

Use Cholesky factors, eigenvalues, or `slogdet`; never compute a raw determinant. If the current precision is singular, use a scientifically defined prior, a reported pseudodeterminant, or a criterion restricted to the identifiable subspace. Numerical jitter used only to complete a factorization is not prior knowledge and must be reported separately.

9.3 Robust or pseudo-Bayesian design

A local design evaluated at a poor estimate can select a poor experiment. Draw bounded samples

$$\phi^{(s)} \sim \mathcal{N}(\hat{\phi}, \mathbf{V}_{\phi}), \quad s = 1, \dots, N_{\text{samp}},$$

and average the D-optimal gain:

$$\overline{\Delta\Psi}_D(\mathbf{u}_c) = \frac{1}{N_{\text{samp}}} \sum_{s=1}^{N_{\text{samp}}} \left\{ \log \det[\mathbf{P}_N + \mathbf{F}_c(\mathbf{u}_c; \phi^{(s)})] - \log \det(\mathbf{P}_N) \right\}.$$

This is computationally more expensive but safer early in the campaign.

9.4 Candidate design space and feasibility

A discrete, reproducible candidate pool is the recommended first implementation. It may be generated from grids, a Latin hypercube, or another space-filling design. Every candidate must respect:

- temperature, pressure, and flow limits;
- allowable pressure drop;
- composition, solubility, and phase constraints;
- catalyst and materials-compatibility limits;
- NMR signal-to-noise and quantitation ranges;
- minimum and maximum useful conversion;
- duplicate and replicate policy.

A simulation failure marks one candidate infeasible; it must not crash the campaign. The ranked table should retain the candidate conditions, criterion value, predicted information gain, minimum eigenvalue, condition number, feasibility flag, and failure reason.

The complete virtual autonomous campaign

10.1 Initialization before autonomy

An autonomous campaign should not begin with an uninformative prior and a blind optimizer. The recommended sequence is:

1. nonreactive tracer RTD and pressure-drop mapping;
2. transfer-line delay and washout characterization;
3. NMR calibration mixtures covering expected composition, acidity, and temperature;
4. four to eight initial kinetic experiments selected by a space-filling design;
5. at least one replicate for noise estimation;
6. at least one holdout condition for early predictive validation;
7. initial parameter estimation and identifiability diagnosis;
8. activation of sequential MBDoE.

10.2 Virtual truth versus inference model

The virtual reactor is an experimental surrogate. It receives a candidate $\mathbf{u}_e$, solves the truth model using hidden ϕ_{true} , applies the observation operator, and adds seeded synthetic noise. The estimator receives only the resulting observations and metadata.

Truth and inference models may initially be identical to validate the algorithm. Later mismatch studies can deliberately make them different:

- truth uses activity-based kinetics; inference uses concentrations;
- truth includes axial dispersion; inference assumes ideal PFR;
- truth includes transfer-line reaction; inference uses only a delay;
- truth contains an additional side reaction.

These extensions test robustness and model discrimination, but they should not bury the reliable core implementation.

10.3 One iteration

At iteration N :

1. append the latest $\mathbf{Y}_e$, labels, covariance, and operating condition to the dataset;
2. fit $\hat{\phi}_N$ using all accumulated experiments;
3. compute residual and validation diagnostics;
4. construct $\mathbf{S}_e$, $\mathbf{S}_{\text{all}}$, $\mathbf{F}_e$, $\mathbf{F}_{\text{data}}$, and $\mathbf{P}_N$;
5. derive $\mathbf{V}_\phi$, $\mathbf{V}_\theta$, and $\mathbf{R}_\theta$;
6. generate or filter the safe candidate pool;
7. calculate $\mathbf{F}_c$ and a design score for every feasible candidate;
8. select $\mathbf{u}_{\text{next}}$;
9. execute the virtual or real experiment;
10. checkpoint all matrices, diagnostics, seeds, and decisions.

10.4 Stopping criteria

Stopping should combine uncertainty, prediction, and experimental value. Suitable conditions are:

- all target relative parameter standard deviations are below thresholds;
- $\det(\mathbf{V}_\theta)$ or a stable log-volume measure is below a target;
- validation RMSE is below a target and no longer improves materially;
- maximum candidate information gain falls below a threshold;
- the maximum experiment count or material budget is reached;
- no safe feasible candidate remains;
- repeated model-diagnostic failures require human review.

Parameter convergence alone is insufficient. A wrong model can converge confidently.

How to prove that spatial autonomy adds value

11.1 The four-strategy benchmark

The central demonstration should compare four strategies under the same truth, noise realizations, initial experiments, candidate pool, safety constraints, and PFR-evaluation budget:

Strategy	Measurement	Experiment selection
1. Outlet, fixed	outlet only	predetermined or space-filling
2. Spatial, fixed	all CPR sampling positions	predetermined or space-filling
3. Outlet, autonomous	outlet only	MBDoE
4. Spatial, autonomous	all CPR sampling positions	MBDoE

This 2×2 design separates two questions:

1. What is gained by spatial measurements?
2. What is gained by adaptive experiment selection?

The expected winner is the spatial-autonomous strategy, but the benchmark must measure rather than assume that result.

11.2 Required performance metrics

After each experiment, record:

- physical-parameter error in retrospective virtual evaluation;
- parameter standard deviations and confidence intervals;
- Fisher eigenvalues, numerical rank, and condition number;
- parameter correlations;
- training weighted residual;
- holdout or external validation RMSE;
- uncertainty in predicted profiles and outlet performance;

- cumulative number of PFR solves;
- failed candidate simulations;
- wall-clock runtime and experimental/material cost.

11.3 Mandatory explanatory figures

The showcase should include:

1. spatial concentration profiles;
2. outlet ambiguity with different internal profiles;
3. sensitivity heatmaps labelled by sampling position, species, and parameter;
4. outlet versus spatial Fisher eigenvalues;
5. parameter-correlation matrices and confidence ellipses;
6. candidate D-optimal score maps or ranked plots;
7. chronological selected-experiment trajectories;
8. parameter estimates and confidence intervals versus iteration;
9. validation RMSE and covariance volume versus experiment count;
10. the four-strategy efficiency comparison.

Layer-2 software architecture

12.1 Contract with Layer 1

Layer 1 remains the authoritative reactor physics. Layer 2 should use a thin adapter rather than rewrite it. The adapter needs to:

- accept an experiment specification and a physical parameter set;
- solve along the primary coordinate z ;
- interpolate the state at configured CPR sampling positions, which are commanded coordinates of one movable orifice rather than fixed hardware ports;
- select measured species in a declared order;
- return predictions with units and labels;
- report solver status, physical violations, and diagnostics.

Layer-1 physics must not be changed unless a verified bug is found. Any unavoidable physics change should be explained and isolated from the Layer-2 implementation.

12.2 Core typed scientific objects

The exact class names may follow repository conventions, but Layer 2 needs objects equivalent to:

MeasurementMatrix

Y_observed	(n_positions, n_species)
Y_predicted	(n_positions, n_species)
y_observed	(n_observations,)
y_predicted	(n_observations,)
covariance	(n_observations, n_observations)
observation_labels	
flattening_order	

SensitivityResult

S_raw	(n_observations, n_parameters)
S_whitened	(n_observations, n_parameters)
parameter_labels	
observation_labels	

```

parameterization
finite_difference_steps
diagnostics

```

InformationResult

```

F_data          (n_parameters, n_parameters)
F_prior         (n_parameters, n_parameters)
P_total         (n_parameters, n_parameters)
covariance_phi
covariance_theta
correlation_theta
eigenvalues, singular_values, rank, condition_number
log_determinant, used_pseudoinverse, diagnostics

```

12.3 Recommended module boundaries

A clean implementation will normally contain:

- **layer1_adapter**: PFR calls, interpolation, species extraction;
- **experiments**: design variables, bounds, units, candidate pools;
- **observations**: **Y**, flattening, labels, covariance;
- **virtual_reactor**: hidden truth and synthetic noise;
- **parameters**: transforms, bounds, units, priors;
- **estimation**: whitening, WNLS/MAP, multistart fitting;
- **sensitivities**: finite differences and diagnostics;
- **information**: FIM, covariance, rank, correlation;
- **design**: D/A/E criteria, robust scores, ranking;
- **campaign**: sequential loop, stopping, checkpoints;
- **validation**: holdouts, residual diagnostics, metrics;
- **plotting**: reusable publication figures;
- **persistence**: JSON/CSV/NPZ and labelled matrices.

No scientifically important logic should exist only inside a notebook.

12.4 Configuration profiles

Use one code path with several configurations:

Fast profile. Few parameters, sampling positions, candidates, and Monte Carlo repetitions for tests and continuous integration.

Demonstration profile. Complete profiles, a scientifically useful candidate pool, several closed-loop iterations, and all showcase figures.

Publication profile. Larger candidate pool, robust parameter sampling, multiple noise realizations, high-resolution output, and complete diagnostics.

All seeds, parameter definitions, sampling-position coordinates, measured species, noise models, bounds, units, and stopping rules belong in validated configuration rather than scattered constants.

Implementation sequence and numerical safeguards

13.1 Phase order

Phase 1: Foundation

- inspect the existing repository and tests;
- implement the Layer-1 adapter;
- define experiment and parameter structures;
- construct labelled observations and position-first flattening;
- add deterministic synthetic noise and dataset persistence.

Phase 2: Inference

- parameter transforms and bounds;
- residual whitening;
- nonlinear estimation;
- finite-difference sensitivities;
- Fisher information, covariance, and identifiability diagnostics.

Phase 3: Experimental design

- reproducible candidate spaces;
- D-, A-, and E-optimal scoring;
- optional robust D-optimality;
- candidate ranking and stopping rules.

Phase 4: Closed loop

- virtual reactor;

- sequential campaign;
- checkpoint/restart;
- holdout validation and complete iteration history.

Phase 5: Showcase

- outlet-versus-spatial ambiguity;
- four-strategy benchmark;
- Monte Carlo repetitions;
- final figures and comparison tables.

Phase 6: Optional scientific extensions

- competing kinetic models and model discrimination;
- concentration versus activity models;
- ideal PFR versus axial-dispersion mismatch;
- reactive transfer-line mismatch.

13.2 Numerical safeguards

Layer 2 must explicitly handle:

- singular or nearly singular information matrices;
- non-positive-definite measurement covariance;
- PFR solver failure and NaN/infinite predictions;
- optimizer evaluations outside physical bounds;
- materially negative concentrations;
- finite-difference perturbations crossing bounds;
- candidates with negligible or complete conversion at every sampling position;
- poor parameter scaling;
- duplicate experiments;
- semidefinite log-determinants;
- failed candidate simulations without campaign failure.

Do not silently clip negative concentrations to zero. Tiny negative values within a configured solver tolerance may be reported and treated numerically; larger values are model or solver failures.

Testing and reproducibility

14.1 Unit tests

At minimum, test:

- parameter transforms and inverse transforms;
- delta-method covariance transformation;
- observation ordering and labels;
- sampling-position interpolation and outlet extraction;
- deterministic noise and covariance construction;
- covariance positive-definiteness checks;
- residual whitening;
- finite-difference sensitivities;
- $\mathbf{F} = \mathbf{J}^\top \mathbf{J}$;
- stacked versus summed Fisher matrices;
- D-, A-, and E-optimal scores and candidate ranking;
- stopping rules;
- checkpoint persistence and reload.

14.2 Analytical validation

For $A \rightarrow B \rightarrow C$:

1. compare numerical and analytical concentration sensitivities;
2. verify the expected matrix dimensions;
3. verify that early positions predominantly inform k_1 ;
4. verify that later positions add information about k_2 ;
5. confirm that adding independent internal rows produces a positive-semidefinite information increment;

6. confirm that a mass-balance-derived species is not counted as independent.

14.3 Integration and regression tests

Integration tests should generate one complete virtual experiment, fit a small parameter subset, execute two closed-loop iterations, run all four strategies on a tiny pool, verify that the selected candidate is feasible, and prove that truth parameters never enter the estimator.

Regression tests should use fixed seeds and realistic numerical tolerances. Exact floating-point equality is usually brittle and scientifically unnecessary.

14.4 Reproducible command-line workflows

Commands should follow the repository conventions but provide workflows equivalent to:

```
python -m scripts.run_outlet_ambiguity_demo \  
    --config configs/virtual_campaign.yaml  
  
python -m scripts.run_virtual_campaign \  
    --strategy spatial_autonomous \  
    --config configs/virtual_campaign.yaml  
  
python -m scripts.run_four_strategy_benchmark \  
    --config configs/benchmark.yaml  
  
python -m scripts.generate_showcase_figures \  
    --results results/latest
```

Each run should print and save its resolved configuration, preserve seeds, report progress, write machine-readable results to a timestamped directory, save figures, and produce an execution summary.

Experimental qualification and scientific risks

15.1 Hydrodynamic qualification

Before fitting kinetics:

1. perform nonreactive step and pulse tracer tests across flow and temperature;
2. estimate RTD mean, variance, skewness, and axial dispersion;
3. map pressure drop and compare with an Ergun estimate;
4. repeat after repacking;
5. vary bead size and packing procedure;
6. test profile sensitivity to sidestream withdrawal;
7. compare ideal-PFR and axial-dispersion predictions.

Accept the packed configuration for kinetic work only when its hydrodynamics are reproducible and the selected model predicts tracer data across the intended design space.

15.2 NMR quantitation risk

Low-field NMR compresses the chemical-shift axis. EGDA, EGMA, EG, AcOH, water, and catalyst effects may create overlap or shifting resonances. A defensible workflow is:

1. acquire high-field reference spectra and Fourier-80 spectra of pure components;
2. prepare calibration mixtures spanning concentration, acidity, and temperature;
3. use constrained line shapes or indirect hard modelling;
4. quantify spectral-fit covariance;
5. validate against gravimetric mixtures or an independent method;
6. reserve direct integration for isolated, demonstrated regions.

15.3 Reactive transfer and sequential sampling

Sampling positions are measured sequentially with the same movable capillary, not simultaneously. Slow drift during a profile scan, capillary repositioning, incomplete washout of the single path, and continued reaction can correlate errors and distort the apparent profile. Record timestamps for motion start and completion, path flushing, stabilization, passage through the NMR-sensitive volume, acquisition, and wash. Change the sampling flow and the flushing volume during commissioning and verify that corrected concentrations at each z_j collapse. If they do not, the observation operator is incomplete.

15.4 Inference risk

Nonlinear kinetic fits may have local minima and apparently small local covariance. Use transformed coordinates, multistart global-to-local optimization, parameter-profile diagnostics, holdout validation, and deliberately varied experimental conditions. The greatest danger is not failure to fit; it is a convincing fit to the wrong effective model.

Required outputs, reporting, and definition of done

16.1 Matrices and metadata saved after every iteration

Save:

- each $\mathbf{Y}_e$ and $\hat{\mathbf{Y}}_e$;
- $\mathbf{y}_{\text{all}}$ and $\hat{\mathbf{y}}_{\text{all}}$;
- raw and whitened residuals;
- Σ_{all} ;
- each $\mathbf{S}_e$, $\mathbf{S}_{\text{all}}$, and whitened sensitivities;
- each $\mathbf{F}_e$, $\mathbf{F}_{\text{data}}$, and $\mathbf{F}_{\text{prior}}$;
- $\mathbf{P}_N$, $\mathbf{V}_\phi$, $\mathbf{V}_\theta$, and $\mathbf{R}_\theta$;
- eigenvalues, singular values, rank, and condition number;
- every candidate $\mathbf{F}_c$ and the ranked candidate table;
- all row/column labels, units, parameterization, seeds, and finite-difference steps.

16.2 Campaign summary

The final report should contain:

- truth parameters only for retrospective virtual evaluation;
- initial guesses and final fitted physical parameters;
- units, errors, standard deviations, and confidence intervals;
- parameter correlation matrix;
- Fisher eigenvalues and condition number;
- selected experiments in chronological order and their criteria;
- validation errors and stopping reason;

- number of PFR evaluations and failed simulations;
- runtime;
- the four-strategy comparison table.

16.3 Definition of done

Layer 2 is complete when:

1. all existing Layer-1 tests still pass;
2. all new tests pass;
3. synthetic spatial datasets can be generated;
4. a configurable kinetic parameter subset can be recovered;
5. outlet-only and spatial information can be compared;
6. MBDoE can select a feasible next condition;
7. a campaign can run without manual intervention;
8. all four strategies can be compared fairly;
9. uncertainty and validation error are tracked after every run;
10. required figures and tables are generated;
11. results are deterministic under a fixed seed;
12. real CPR–NMR measurements can later replace the virtual observations without rewriting estimation and design logic;
13. exact reproduction commands are documented.

Recommended development roadmap

17.1 Near-term scientific milestones

Milestone 1: analytical core. Reproduce the complete $A \rightarrow B \rightarrow C$ example, including labelled $\mathbf{Y}$, $\mathbf{y}$, $\mathbf{S}$, $\mathbf{F}$, covariance, outlet extraction, and candidate scoring.

Milestone 2: Layer-1 adapter. Generate position-resolved EGDA predictions from the existing spatial PFR without modifying its physics.

Milestone 3: virtual estimation. Recover a small, well-identified subset such as two reference-temperature rate constants before attempting all Arrhenius and reverse parameters.

Milestone 4: virtual autonomy. Complete a deterministic closed-loop campaign and show that selected conditions reduce validation error and uncertainty.

Milestone 5: four-strategy evidence. Demonstrate separately the value of spatial profiles and adaptive selection.

Milestone 6: hardware qualification. Complete RTD, pressure-drop, transfer-line, and NMR quantitation studies.

Milestone 7: real proof of concept. Replace synthetic $\mathbf{Y}_e$ with measured CPR–NMR profiles while retaining the same Layer-2 interfaces.

17.2 Scientific escalation order

Fit complexity only as the data can support it:

1. one-step irreversible or pseudo-first-order commissioning model;
2. two-step irreversible analytical validation;
3. concentration-based reversible EGDA model at one temperature;
4. temperature-dependent forward and reverse kinetics;
5. composition-dependent/activity corrections;
6. hydrodynamic and transfer mismatch;
7. competing mechanisms and model discrimination.

This order does not deny the real complexity. It prevents complexity from being confused with scientific completeness before the basic information machinery is verified.

Final interpretation

The CPR does not create a new type of kinetic parameter. Its advantage is that one steady operating condition creates sensitivity rows at several reaction stages. Early positions reveal reactant disappearance and first-step kinetics. Intermediate positions capture formation and consumption of EGMA. Later positions reveal the second step, equilibrium approach, and reverse effects. These rows improve the rank and conditioning of the information matrix when they point in parameter directions that are missing from the outlet.

The Fourier 80 makes those rows chemically meaningful by estimating several species from each withdrawn mixture. The observation model and covariance determine how much of that apparent information is trustworthy. The Layer-1 model maps parameters and operating conditions to spatial concentrations. Layer 2 then turns concentrations into residuals, sensitivities, information, uncertainty, and a ranked next experiment.

The central idea in one sentence

The self-driving laboratory repeatedly chooses the temperature, flow, and composition expected to create a future spatial sensitivity matrix whose Fisher-information contribution most strongly reduces the uncertainty that remains in the kinetic model.

That matrix-based process is the actual brain of the platform. Automation of pumps, the capillary positioner, sampling flow, and spectra is necessary, but the scientific value comes from the disciplined connection between physical state, measured data, uncertainty, parameter identifiability, predictive validation, and the next decision.

Notation and dimension summary

Symbol	Dimension	Meaning
$\mathbf{Y}_e$	$N_z^{(e)} \times N_s^{(e)}$	structured observed spatial profile
$\hat{\mathbf{Y}}_e$	$N_z^{(e)} \times N_s^{(e)}$	model-predicted spatial profile
$\mathbf{y}_e$	m_e	position-first observation vector
$\hat{\mathbf{y}}_e$	m_e	prediction vector in identical order
$\Sigma_{y,e}$	$m_e \times m_e$	observation error covariance
$\mathbf{d}_e$	m_e	raw residual
$\mathbf{r}_e$	m_e	whitened residual
$\boldsymbol{\theta}$	N_p	physical fitted parameters
$\boldsymbol{\phi}$	N_p	transformed internal parameters
$\mathbf{S}_e$	$m_e \times N_p$	raw prediction sensitivity
$\mathbf{J}_e$	$m_e \times N_p$	whitened sensitivity
$\mathbf{F}_e$	$N_p \times N_p$	information from experiment e
$\mathbf{F}_{\text{data}}$	$N_p \times N_p$	accumulated data information
$\mathbf{F}_{\text{prior}}$	$N_p \times N_p$	prior precision
$\mathbf{P}_N$	$N_p \times N_p$	total local precision
$\mathbf{V}_\phi$	$N_p \times N_p$	local covariance in internal coordinates
$\mathbf{V}_\theta$	$N_p \times N_p$	local covariance in physical coordinates
$\mathbf{R}_\theta$	$N_p \times N_p$	physical-parameter correlation
$\mathbf{u}_e$	N_u	controllable condition for experiment e

Non-negotiable scientific rules

1. Position z is the primary PFR coordinate; residence time is derived.
2. Sensitivities are derivatives of predictions with respect to fitted parameters.
3. Covariance diagonals are variances.
4. Equal independent noise is an option, not a default truth.
5. Fisher matrices may be added only for block-independent errors.
6. Calculated mass-balance quantities are not independent measurements.
7. Parameterization must be consistent in fitting, sensitivities, and design.
8. Stable linear algebra replaces explicit inverses and raw determinants.
9. Hidden truth is unavailable to fitting and experiment selection.
10. Training fit alone cannot establish predictive success.
11. Small local covariance cannot establish model correctness.
12. Layer-1 physics remains unchanged unless a verified bug is isolated.

Selected references and further reading

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End note

This compendium is organized as a single argument: qualify the physical system, define the observation, learn parameters with honest uncertainty, convert prediction sensitivities into information, validate the model, and only then let the information structure choose the next experiment.